

IBM BOB 2.0 HACKATHON · LABLAB.AI · SEP 2026

Honest Postmortems from Repository Evidence

From the fix commit to a full incident postmortem — which commit introduced the bug, the timeline, the root cause, the prevention plan. Every claim linked to evidence. Nothing invented.

● every claim linked to evidence · ● grounded, some claims unreferenced · ● no evidence (declared)

Incident Postmortem Generator · solo builder · github.com/el-informatico/postmortem-generator · Apache-2.0

PROBLEM

The postmortem nobody has time to write

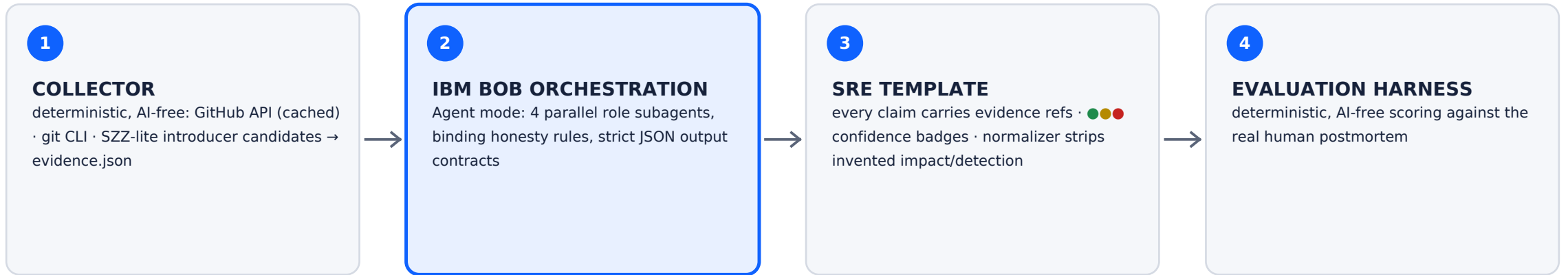
- An incident ends. The fix is merged. And the postmortem — the document where the team actually learns — never gets written.
- Writing one costs a senior engineer ~90 minutes, weeks after the pager went quiet. So most postmortems simply don't exist.
- Incident-management tools (Rootly, PagerDuty, incident.io, FireHydrant) generate postmortems from the managed incident stack: alerts, Slack threads, timelines.
- Most bug fixes leave none of that behind. There is no incident record to mine — but the repository remembers everything.
- The evidence for the postmortem already exists, untouched, in the issue, the fix commit, and the commit that introduced the bug.

SOLUTION

Repo-first postmortems, honesty enforced

- Start from the repository, not the incident stack: give it a repo, the issue behind a bug, and the SHA of the fix — get a full postmortem out.
- Introducer discovery via **SZZ-lite**: blame, pickaxe and issue linkage trace the fix's deleted lines back to the commit that introduced the bug.
- A deterministic collector (GitHub API + git CLI, zero AI) builds the evidence; parallel IBM Bob role agents analyze it; an SRE synthesizer writes the report.
- Every claim carries evidence refs. Every section carries a confidence badge. Sections git cannot ground are declared ● No evidence — the harness FAILS any run that invents them.
- A complement to telemetry-centric AIOps (Instana, Cloud Pak): it closes the post-fix loop those stacks never see. Honesty is enforced, not promised.

Four separable pieces — not a prompt wrapper



- Ships as `bob postmortem --repo ... --issue ... --fix-sha ...` (Bob Shell slash command) and a post-merge CI step; an offline deterministic fallback runs wherever no key is configured.
- The prompt is the last 10% — the product is the pipeline.

How IBM Bob 2.0 was used

- **Agent mode, custom role modes** (.bob/): Commit Archaeologist · Document Analyst · Code Reviewer · SRE Synthesizer — parallel subagents, each seeing ONLY its slice of the evidence.
- **Binding honesty rules + output contracts per role:** JSON findings only, refs restricted to the provided evidence ref-id list; a normalizer discards any invented impact or detection text.
- **Document understanding over real issue threads:** curl issue #4907 is a PR that was closed *unmerged* — a tool that assumes "closed means merged" gets this case wrong.
- **Full session evidence trail:** per-task consumption summaries (task ID, tokens, cost) captured in bob_sessions/ — see the Evidence slide.
- **Measured cost, not estimates:** \$0.07 across the four role sessions (\$0.10 including the access smoke test) — the 40-Bobcoin sprint budget essentially untouched.
- Scope, stated plainly: deterministic core is pre-sprint work (open, dated git history); the IBM Bob integration layer was built in the sprint window.

Real numbers against the human postmortem — no spin

Metric (vs Daniel Stenberg's hand-written postmortem)	Deterministic floor	IBM Bob 4-role run
introducer top-1 / top-5	✓ exact (4a4b63daaa)	✓ exact (4a4b63daaa)
honesty check (impact & detection red)	✓	✓
claim linkage	1.00 (21/21)	1.00 (26/26)
timeline recall / precision	0.75 / 1.00	0.75 / 1.00
action-item overlap	0.33	0.22
root-cause match	0.29	0.14
line agreement (human prose)	0.00	0.01
wall clock	<1 s	56 s (human: ~90 min)

- Repo-derivable metrics hold at ceiling under agent mode, with a richer report: 26 linked claims vs 21, every timeline entry precise.
- Narrative overlap did **not** beat the deterministic floor yet — reported as-is. That open gap is the declared next target, not a hidden number.

Every session has a receipt

Bob session (custom role mode)	Task ID	Context tokens	Cost
Commit Archaeologist	13e68f09bab1b1b5	8,676	\$0.017
Document Analyst	2d8266ccd2f2bb4d	5,854	\$0.012
Code Reviewer	900fb3e2e9a5ac1f	7,893	\$0.016
SRE Synthesizer	d32e706198b8c736	14,695	\$0.029
Access smoke test	f404dad9621609031	—	\$0.022

- Screenshots of each task-session summary + exported task histories live in bob_sessions/ in the repo — judges can audit every claim on this deck.
- The scored case is real: curl CVE-2023-38545 — introducer 4a4b63daaa (2020-02-14) → fix fb4415d8aee6 (2023-10-11): the bug's origin story, **1,335 days** before the fix.
- Evaluation is deterministic and AI-free — 175 offline tests, byte-reproducible scoring.

Where it doesn't work (yet)

- **One case scored end-to-end** (curl). Two further real cases exercise the edges of the approach:
- **log4j CVE-2021-44228** — introducer top-1 ✗: SZZ-lite honestly blames later JNDI refactors, not the 2013 component introduction. The floor's known blind spot, visible by design.
- **GitLab 2017 database outage** — an ops incident with no fix commit: degrades to an issues-only reconstruction and says so (resolution: ● No evidence — no fix commit).
- **Attribution is candidates, not proof**: every attribution carries its score and badge; a wrong guess surfaces in yellow or red — never buried under confident prose.
- Agent-mode narrative scores still sit below the deterministic floor (see Metrics) — shown, not spun.

SEE IT RUN

Demo & repository

- **Demo video (4:27):** the full pipeline over the real curl CVE — interactive timeline UI, click-through from any claim to its evidence, side-by-side vs the human postmortem.
- **Repo:** github.com/el-informatico/postmortem-generator (public, Apache-2.0)
- One command: `bob postmortem --repo curl/curl --issue 4907 --fix-sha fb4415d8aee6`
- 3 real cases · 175 offline tests · CI workflow · deterministic, byte-reproducible evaluation
- IBM Bob 2.0 is the orchestration layer. Your repository is the evidence. The postmortem nobody had time to write — in seconds, with receipts.